

JavaScript Key Learnings — Question 6

OJT Assessment Prep — window Properties, Viewport, Redirect & Logic Operators

1. window.innerWidth — Reading Viewport Size

The **window** object represents the browser. **innerWidth** and **innerHeight** give you the current visible area in pixels.

```
window.innerWidth // → current viewport width in pixels (e.g. 1440)
window.innerHeight // → current viewport height in pixels (e.g. 900)

// Use it to detect screen size:
if (window.innerWidth < 1024) {
  // mobile or tablet
} else {
  // desktop
}
```

■ Common breakpoints: mobile < 768px, tablet < 1024px, desktop >= 1024px.

2. Setting CSS Class via JavaScript — .className

You can assign a CSS class to a dynamically created element using **.className**:

```
const div = document.createElement("div");
div.className = "warning";
// same as writing: <div class="warning"> in HTML

// You can also add multiple classes:
div.className = "warning active large";
```

3. window.location.href — Browser Redirect

window.location.href controls the current URL. Setting it navigates the browser to a new page.

```
// Read current URL
console.log(window.location.href); // → "https://current-page.com"

// Redirect to a new URL
window.location.href = "https://mobile.company.com";
// browser immediately navigates there
```

Used with **setTimeout** to redirect after a delay — exactly what this question does.

4. The ! Operator Trap — Logic Bug

The most common logic mistake in this question — misplacing the ! (NOT) operator.

■ Wrong — ! applied to the number, not the comparison:

```
if (!window.innerWidth < 1024)
// Step 1: !window.innerWidth → !884 → !truthy → false
// Step 2: false < 1024 → 0 < 1024 → always true!
// Result: warning shows on ALL screens – broken!
```

■ **Correct** — just use the comparison directly:

```
if (window.innerWidth < 1024)
// 884 < 1024 → true → show warning
// 1440 < 1024 → false → do nothing
// Clean, simple, correct!
```

■■ **Rule:** Never put ! before a number — it converts it to boolean first, breaking your comparison.

5. Combining Everything — The Full Pattern

This question combines concepts from all previous questions:

```
function enforceDesktopView() {
  if (window.innerWidth < 1024) { // check viewport
    let div = document.createElement("div"); // create element
    div.className = "warning"; // set CSS class
    div.textContent =
      "Please switch to a desktop device."
      + " You will be redirected in 5 seconds.";
    document.body.append(div); // add to page
    setTimeout(() => {
      window.location.href =
        "https://mobile.company.com"; // redirect after 5s
    }, 5000);
  }
}

enforceDesktopView();
```

6. Debugging viewport Width — The DevTools Trap

A very common confusion during testing — DevTools shrinks your reported viewport width!

Situation	window.innerWidth reports
Full screen browser, no DevTools	~1440px (desktop) ■
DevTools open on the side	~558px (looks mobile!) ■■
DevTools responsive mode on	Whatever you set (e.g. 375px) ■■
Browser window not maximized	Less than full screen width ■■

■ **Debug tip:** Always run `console.log(window.innerWidth)` first to see what JS actually reads.

```
// Add this to debug:
function enforceDesktopView() {
  console.log("Width is:", window.innerWidth); // check this first!
  if (window.innerWidth < 1024) { ... }
```

```
}
```

7. Useful window Properties — Quick Reference

Property	What it gives you
<code>window.innerWidth</code>	Viewport width in pixels
<code>window.innerHeight</code>	Viewport height in pixels
<code>window.location.href</code>	Current URL — set it to redirect
<code>window.location.reload()</code>	Refresh the current page
<code>window.alert("msg")</code>	Show a browser alert box
<code>window.innerWidth</code>	Viewport width in pixels

■ ■ Full Flow Summary

1. Call `enforceDesktopView()`
2. Read `window.innerWidth`
3. if `width < 1024`:
 - a. `createElement("div")`
 - b. `div.className = "warning"`
 - c. `div.textContent = "Please switch..."`
 - d. `document.body.append(div)`
 - e. `setTimeout 5000ms → window.location.href = redirect URL`
4. if `width >= 1024`: do nothing — desktop view is fine